Exercises 8.5

Irreducible Quadratic Factors

In Exercises 21–32, express the integrand as a sum of partial fractions and evaluate the integrals.

30. $\displaystyle\int \frac{x^2 + x}{x^4 - 3x^2 - 4}\,dx$

Improper Fractions

In Exercises 33–38, perform long division on the integrand, write the proper fraction as a sum of partial fractions, and then evaluate the integral.

38. $\displaystyle\int \frac{2y^4}{y^3 - y^2 + y - 1}\,dy$

Evaluating Integrals

Evaluate the integrals in Exercises 39–50.

42. $\displaystyle\int \frac{\sin\theta\,d\theta}{\cos^2\theta + \cos\theta - 2}$

46. $\displaystyle\int \frac{1}{(x^{1/3} - 1)\sqrt{x}}\,dx$

(*Hint:* Let $x = u^6$.)

50. $\displaystyle\int \frac{1}{x^6(x^5+4)}\,dx$

41. $\displaystyle\int_0^{\pi} \frac{dt}{\sqrt{t}+\sin t}$

Exercises 8.8

Testing for Convergence

In Exercises 35–64, use integration, the Direct Comparison Test, or the Limit Comparison Test to test the integrals for convergence. If more than one method applies, use whatever method you prefer.

39. $\displaystyle\int_0^{\ln 2} x^{-2}e^{-1/x}\,dx$

55. $\displaystyle\int_{\pi}^{\infty} \frac{2+\cos x}{x}\,dx$

63. $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{\sqrt{x^4 + 1}}$

Theory and Examples

65. Find the values of p for which each integral converges.

a. $\displaystyle\int_1^2 \frac{dx}{x(\ln x)^p}$

b. $\displaystyle\int_2^{\infty} \frac{dx}{x(\ln x)^p}$

66. $\int_{-\infty}^{\infty} f(x)\,dx$ **may not equal** $\lim_{b\to\infty} \int_{-b}^{b} f(x)\,dx$ Show that

$$\int_0^{\infty} \frac{2x\,dx}{x^2+1}$$

diverges and hence that

$$\int_{-\infty}^{\infty} \frac{2x\,dx}{x^2+1}$$

diverges. Then show that

$$\lim_{b\to\infty} \int_{-b}^{b} \frac{2x\,dx}{x^2+1} = 0.$$

71. Find the area of the region that lies between the curves $y = \sec x$ and $y = \tan x$ from $x = 0$ to $x = \pi/2$.

(Ex.71 不用做！)

72. The region in Exercise 71 is revolved about the x-axis to generate a solid.

a. Find the volume of the solid.

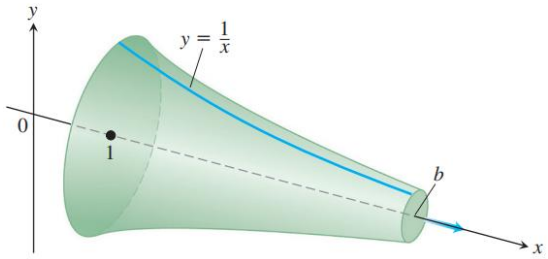
b. Show that the inner and outer surfaces of the solid have infinite area.

76. The infinite paint can or Gabriel's horn As Example 3 shows, the integral $\int_1^\infty (dx/x)$ diverges. This means that the integral

$$\int_1^\infty 2\pi \frac{1}{x}\sqrt{1+\frac{1}{x^4}}\,dx,$$

which measures the *surface area* of the solid of revolution traced out by revolving the curve $y = 1/x, 1 \le x$, about the x-axis, diverges also. By comparing the two integrals, we see that, for every finite value $b > 1$,

$$\int_1^b 2\pi \frac{1}{x}\sqrt{1+\frac{1}{x^4}}\,dx > 2\pi\int_1^b \frac{1}{x}\,dx.$$



However, the integral

$$\int_1^\infty \pi\left(\frac{1}{x}\right)^2 dx$$

for the *volume* of the solid converges.

a. Calculate it.

b. This solid of revolution is sometimes described as a can that does not hold enough paint to cover its own interior. Think about that for a moment. It is common sense that a finite amount of paint cannot cover an infinite surface. But if we fill the horn with paint (a finite amount), then we *will* have covered an infinite surface. Explain the apparent contradiction.

80. Show that if $f(x)$ is integrable on every interval of real numbers and a and b are real numbers with $a < b$, then

a. $\int_{-\infty}^a f(x)\,dx$ and $\int_a^\infty f(x)\,dx$ both converge if and only if $\int_{-\infty}^b f(x)\,dx$ and $\int_b^\infty f(x)\,dx$ both converge.

b. $\int_{-\infty}^a f(x)\,dx + \int_a^\infty f(x)\,dx = \int_{-\infty}^b f(x)\,dx + \int_b^\infty f(x)\,dx$ when the integrals involved converge.

Chapter 8 Practice Exercises

Assorted Integrations

Evaluate the integrals in Exercises 69–116. The integrals are listed in random order so you need to decide which integration technique to use.

93. $\int e^{\ln\sqrt{x}}\,dx$

103. $\int \sqrt{x}\cdot\sqrt{1+\sqrt{x}}\,dx$

109. $\int \frac{x^{\ln x}\ln x}{x}\,dx$

113. a. Show that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$.

b. Use part (a) to evaluate

$$\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,dx.$$

115. $\displaystyle\int \frac{\sin^2 x}{1 + \sin^2 x}\,dx$

Chapter 8 Additional and Advanced Exercises

Evaluating Integrals

Evaluate the integrals in Exercises 1–6.

2. $\displaystyle\int \frac{dx}{x(x + 1)(x + 2)\cdots(x + m)}$

Evaluate the limits in Exercises 7 and 8.

8. $\displaystyle\lim_{x\to 0^+} x\int_x^1 \frac{\cos t}{t^2}\,dt$

Applications

11. Finding arc length Find the length of the curve

$$y = \int_0^x \sqrt{\cos 2t}\, dt, \quad 0 \le x \le \pi/4.$$

26. Length of a curve Find the length of the curve

$$y = \int_1^x \sqrt{\sqrt{t} - 1}\, dt, \qquad 1 \le x \le 16.$$

27. For what value or values of a does

$$\int_1^\infty \left(\frac{ax}{x^2 + 1} - \frac{1}{2x} \right) dx$$

converge? Evaluate the corresponding integral(s).

\

31. Integrating the square of the derivative If f is continuously differentiable on $[0, 1]$ and $f(1) = f(0) = -1/6$, prove that

$$\int_0^1 (f'(x))^2\,dx \geq 2\int_0^1 f(x)\,dx + \frac{1}{4}.$$

Hint: Consider the inequality $0 \leq \displaystyle\int_0^1 \left(f'(x) + x - \frac{1}{2}\right)^2 dx$.

Source: Mathematics Magazine, vol. 84, no. 4, Oct. 2011.

32. (*Continuation of Exercise 31.*) If f is continuously differentiable on $[0, a]$ for $a > 0$, and $f(a) = f(0) = b$, prove that

$$\int_0^a (f'(x))^2\,dx \geq 2\int_0^a f(x)\,dx - \left(2ab + \frac{a^3}{12}\right).$$

Hint: Consider the inequality $0 \leq \displaystyle\int_0^a \left(f'(x) + x - \frac{a}{2}\right)^2 dx$.

Source: Mathematics Magazine, vol. 84, no. 4, Oct. 2011.

Use the substitutions in Equations (1)–(4) to evaluate the integrals in Exercises 41–48. Integrals like these arise in calculating the average angular velocity of the output shaft of a universal joint when the input and output shafts are not aligned.

42. $\displaystyle\int \frac{dx}{1 + \sin x + \cos x}$

51. If n is a nonnegative integer, $\Gamma(n + 1) = n!$

a. Show that $\Gamma(1) = 1$.

b. Then apply integration by parts to the integral for $\Gamma(x + 1)$ to show that $\Gamma(x + 1) = x\Gamma(x)$. This gives

$$\begin{aligned}
\Gamma(2) &= 1\Gamma(1) = 1 \\
\Gamma(3) &= 2\Gamma(2) = 2 \\
\Gamma(4) &= 3\Gamma(3) = 6 \\
&\vdots \\
\Gamma(n + 1) &= n\,\Gamma(n) = n! \qquad (5)
\end{aligned}$$

c. Use mathematical induction to verify Equation (5) for every nonnegative integer n.

52. Stirling's formula Scottish mathematician James Stirling (1692–1770) showed that

$$\lim_{x\to\infty}\left(\frac{e}{x}\right)^x\sqrt{\frac{x}{2\pi}}\,\Gamma(x) = 1,$$

so, for large x,

$$\Gamma(x) = \left(\frac{x}{e}\right)^x\sqrt{\frac{2\pi}{x}}\,(1+\epsilon(x)), \qquad \epsilon(x)\to 0 \text{ as } x\to\infty. \tag{6}$$

Dropping $\epsilon(x)$ leads to the approximation

$$\Gamma(x) \approx \left(\frac{x}{e}\right)^x\sqrt{\frac{2\pi}{x}} \qquad \textbf{(Stirling's formula)}. \tag{7}$$

a. Stirling's approximation for *n*! Use Equation (7) and the fact that $n! = n\Gamma(n)$ to show that

$$n! \approx \left(\frac{n}{e}\right)^n\sqrt{2n\pi} \qquad \textbf{(Stirling's approximation)}. \tag{8}$$

As you will see if you do Exercise 104 in Section 10.1, Equation (8) leads to the approximation

$$\sqrt[n]{n!} \approx \frac{n}{e}. \tag{9}$$

T **b.** Compare your calculator's value for $n!$ with the value given by Stirling's approximation for $n = 10, 20, 30, \ldots$, as far as your calculator can go.

T **c.** A refinement of Equation (6) gives

$$\Gamma(x) = \left(\frac{x}{e}\right)^x\sqrt{\frac{2\pi}{x}}\,e^{1/(12x)}(1+\epsilon(x))$$

or

$$\Gamma(x) \approx \left(\frac{x}{e}\right)^x\sqrt{\frac{2\pi}{x}}\,e^{1/(12x)},$$

which tells us that

$$n! \approx \left(\frac{n}{e}\right)^n\sqrt{2n\pi}\,e^{1/(12n)}. \tag{10}$$

Compare the values given for 10! by your calculator, Stirling's approximation, and Equation (10).